

Data & AI

8. Your Own Image Classifier

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Last Time

A classifier trained on somebody else's data

- 784 numbers in, 10 numbers out
- 50,890 parameters, trained in about a second
- Training accuracy 99.2%, test accuracy 95.0%

Today

- Assignment 1: an image classifier on images you collect
- Git, GitHub and GitHub Pages, because that is how you hand it in
- Google Colab, for those without a fast computer

What you will build

The Demo

espresso_cup · paper_coffee_cup · wine_glass

3 classes, 32x32 colour input, 393,731 parameters.

self test passed: the browser agrees with Python (largest gap 0.0000)

Webcam

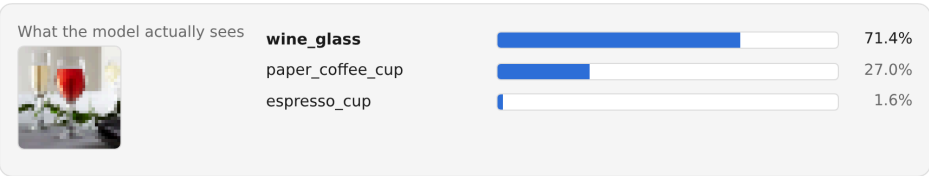
Upload a picture

Draw it

Choose File0025.jpg

Pick a photo from your own phone or computer.





Built for MAS1004 Data & AI, Sogang University. The model is a small neural network trained on images collected by the student. It runs in JavaScript inside this page, so it works with no server at all. The forward pass is in `app.js` and is about forty lines long.

Three Ways to Feed It a Picture

- Your webcam
- A file you upload
- Something you draw with the mouse

It Runs Inside the Page

- No server, no Python, no installing anything
- The address keeps working after you close your laptop
- Nobody's photos are sent anywhere

Assignment 1

Train an image classifier on images you collect yourself

- 3 to 5 categories of your own choosing
- Due Friday 16 October, 22:00
- Everything at github.com/jdasam/mas1004-assignment1

The assignment

The Six Problems

-
- 1 Choose your categories and collect the images
 - 2 Turn your folder into numbers
 - 3 Train it and report the first result
 - 4 Clean your data and train again
 - 5 Test it on photographs you took yourself
 - 6 Put the demo on the web

Two Reports

	Who writes it	How long
Short	You, by hand	One page
Long	You and your agent	As long as you like

The Short Report

- One page. Written by you, with no language model at any stage
- In Korean, if Korean is your first language
- What it does, its worst mistake, where your agent was wrong, what you would change

The Long Report

- Write it with your agent. That is what it is for
- Every number, every table, every figure, the whole checker output
- It will be read by a program, and by you while you write the short one

What You Are Given

- The image downloader
- The web page, with the webcam, upload and drawing already in it
- The export script and the checker
- A test for every function you have to write

What You Write

File	Functions
src/data.py	load_folder, split_train_test
src/train.py	build_model, train
src/evaluate.py	predict_logits, accuracy, confusion_matrix
src/evaluate.py	worst_examples

- Each one already has a description of exactly what goes in and what comes out

How You Know It Worked

```
pytest tests/test_data.py
```

- Red until you write it, green when it is right
- Paste the whole failure text to your agent, not your summary of it

The Plan Is Written Out for You

PLAN.md, sixteen steps, each with a way to check it

- Give it to your agent one step at a time, not all at once
- Read it first. Writing one of these is part of what you are learning

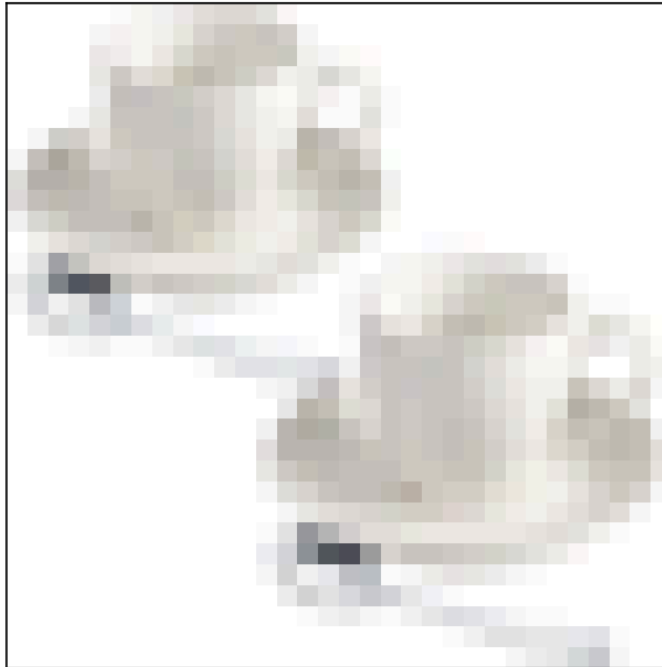
Choosing your categories

What the Model Actually Sees

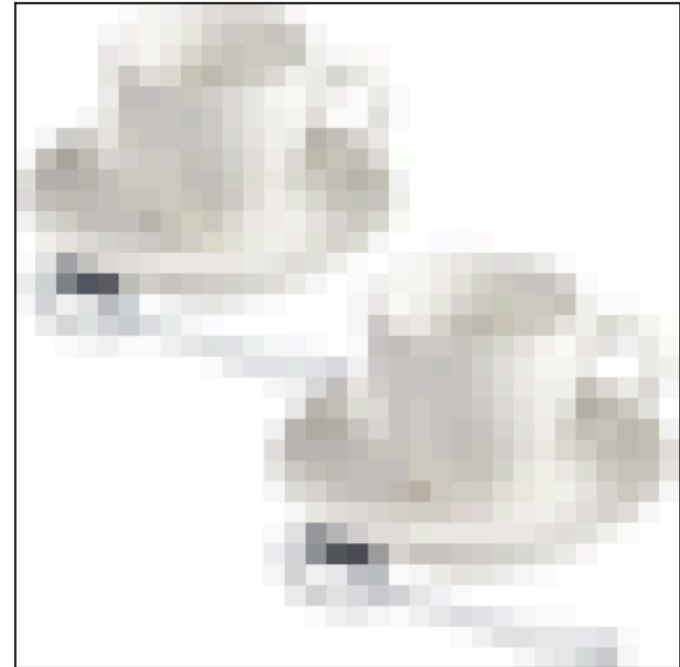
the photo
1000 x 1000 pixels



what you train on
32 x 32 pixels



the same 32 x 32,
blown up so you can see it



Categories That Work

- Told apart by colour and overall shape
- Three to five of them
- At least 50 images each, 150 is better

Pick Something You Care About

- The grade is in the explanation, not the accuracy
- You will look at these images one by one in Problem 4
- Choose something you would still find interesting at 60%

Downloading the Images

```
python src/collect.py --classes "espresso cup,wine glass,paper cup" --n 150
```

- About 70 seconds per category
- Search phrases, not category names: change the phrase if a category comes back empty

Your Test Set Is Not from the Internet

At least 5 photographs per category, taken by you

- Problem 5 measures the model on those, not on downloaded images
- It will do much worse on yours. Explaining that gap is the assignment

Git, GitHub, GitHub Pages

Why You Need These Today

- You hand in a repository address, not a zip file
- Your demo has to be on the web, and GitHub puts it there for free

Git

A program that records every version of your files

- Every save is kept, with a message saying what changed
- You can go back to any of them

GitHub

A website that holds git repositories

The screenshot shows the GitHub interface for the repository `jdasm/mas1004`. The top navigation bar includes links for Platform, Solutions, Resources, Open Source, Enterprise, and Pricing, along with search, sign in, and sign up buttons. The repository header shows the name `jdasm/mas1004` as public, with 10 stars and 1 fork. The main content area displays the repository's file structure and commit history.

File	Description	Time
<code>data</code>	Add MNIST subset for lecture 7	last week
<code>demos</code>	Publish demo: lecture05-neuron-patch	2 weeks ago
<code>slide_decks</code>	Update lecture 7 slide deck	last week
<code>.gitignore</code>	2026 브랜치 시작: 이전 연도 과제와 라이브 코딩 자료를 빼고 ...	last month
<code>.nojekyll</code>	Publish demo: lecture03-regression-search	3 weeks ago
<code>README.md</code>	Add lecture 6 demo links	2 weeks ago

The README section is titled **mas1004** and describes the course materials for Data & AI at Sogang University. It mentions that lecture slides are in the `slide_decks/` directory and that interactive demos are in the `demos/` directory.

About
Data & AI at Sogang University
Readme
Activity
10 stars
1 watching
1 fork
Report repository

Releases
No releases published

Contributors 2
`jdasm` Dasaem Jeong
`claude` Claude

Getting Your Own Copy

1. Make an account at github.com
2. Open github.com/jdasam/mas1004-assignment1
3. Press "Use this template", then "Create a new repository"
4. Keep it public

Three Commands

```
git clone https://github.com/<you>/<your repo>.git  
git add -A  
git commit -m "what I changed"  
git push
```

- Your coding agent can run all of these for you
- Push often. An unpushed commit is not handed in

GitHub Pages

Settings, then Pages, then Branch: main and Folder: /docs

- Your demo appears at `https://<you>.github.io/<your repo>/`
- It takes a minute or two the first time

Google Colab

A Computer in a Data Centre, for Free

- Nothing to install
- Notebook at `assignment1_colab.ipynb` in the repository

Everything You Save There Is Deleted

So keep the work in your Google Drive instead

```
from google.colab import drive  
drive.mount('/content/drive')
```

- The notebook then works inside My Drive/mas1004/assignment1

Turning On the GPU

Runtime, Change runtime type, T4 GPU, Save

- Do it before you run anything. The session restarts
- You do not need it for this assignment. Your model trains on a CPU in seconds

Now, in class

What to Get Done Before You Leave

1. A GitHub account
2. Your own copy of the repository, from the template
3. Your categories, written down
4. `collect.py` running

Before Thursday

- Take photographs of your own categories. At least 5 each
- Read `PLAN.md` and `README.md` all the way through
- Problems 1 and 2 finished

Coming Up

- Thursday: the training loop, opened up in PyTorch
- Then: images with more than one colour channel, and why 32 by 32 hurts

What would you want it to tell apart?